Delay as a signal of deliberation*

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ABSTRACT

A respondent who is asked a question may either wait until a well-considered answer comes to mind or fabricate one on the spot. Because fabrication is fast, a prompt reply is suspected of being fabricated, and the respondent may delay a well-considered answer to avoid this suspicion. We study a continuous-time model in which the arrival rate of a well-considered answer varies over time and respondents may become impatient, and hence tempted to fabricate, as time passes. Intentional delay arises when the arrival rate is strictly increasing and the respondent is sufficiently patient, and it never arises when the arrival rate is constant or decreasing; only impatient respondents fabricate. As long as potential fabricators are abundant, the equilibrium suspicion attached to a reply at any instant depends neither on the initial share of impatient respondents nor on the rate at which respondents become impatient. Policies that reduce the supply of fabricators therefore leave suspicion, and the delay of honest respondents, unchanged until fabricators become scarce; they only reshape the timing of replies. Similarly, a more lenient questioner faces more suspicion and more fabrication, but investigates any given reply with the same probability.

Keywords: Strategic delay; fabrication; inspection games; signaling through timing; continuous time game

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1. Introduction

Fabricating a plausible answer has never been cheaper. Whether a question comes from a survey, an employer, a client, or a teacher, the respondent can now obtain a fluent reply from a large language model in seconds, and many do: Veselovsky, Horta Ribeiro, and West (2023) estimate that between a third and nearly half of the crowdsourced text-production tasks they study were completed with the help of such models. Because a fabricated reply is increasingly hard to distinguish from a considered one by its content alone, questioners turn to when it arrives. Survey researchers routinely flag and discard responses that come in "too fast" (Greszki, Meyer, and Schoen, 2015), and experimental evidence confirms that observers treat response time as a cue to the quality of the underlying thought: Van de Calseyde, Keren, and Zeelenberg (2014) find that slower decisions are read as more deliberate, and Ziano (2018) documents that slow responders are attributed higher status. Speed has become a proxy for fabrication.

This creates a perverse incentive for honest respondents. Someone who has genuinely thought of an answer may hold it back, because a quick reply risks being suspected of fabrication while a measured one reads as considered. The pause is then not deliberation but its imitation. The phenomenon is older than generative AI. Central banks schedule the release of statements rather than publish as soon as drafts are ready, expert witnesses rarely answer in a single breath, and consultants are advised to let a beat pass before responding to a client. Cheap fabrication, however, has made it pervasive. The common thread is that the timing of a reply carries information about whether the reply is well-considered, and respondents have an incentive to manage that inference. A natural response is to fight fabrication directly, by detecting or prohibiting the tools that make it easy. One of our results suggests that this alone does not relieve honest respondents of the need to delay.

This paper studies a continuous-time model of that incentive. A respondent is asked a question and, over time, randomly thinks of a well-considered answer at an instantaneous rate $\theta(t)$. Before such an answer arrives, she can fabricate one and reply immediately; after it arrives, she can reply immediately or delay. The questioner does not observe whether a given reply is well-considered, but draws an inference from the time at which it is delivered. The respondent's patience is private information, and a patient respondent may become impatient as time passes: she may tire of

the question and be tempted to let a model answer it for her. The questioner may investigate a reply, and he does so when his posterior that it was fabricated, refined by a noisy signal, is high enough. An investigation is costly even for an honest respondent, who may be falsely charged. This false-charge aversion makes suspicion costly for honest and dishonest respondents alike.

Three results follow. First, the shape of θ governs intentional delay of a well-considered answer. When $\theta(t)$ is strictly increasing, waiting genuinely makes the arrival of a good answer more likely, and a late reply is a priori more credible. In that case a sufficiently patient respondent delays an answer she already has. When θ is constant or decreasing, no such delay occurs. Second, fabrication is the exclusive behavior of impatient respondents: patient respondents always wait for a well-considered answer. Third, and most strikingly, as long as enough impatient respondents remain to fabricate, the equilibrium suspicion that a reply is fabricated (the questioner's posterior at any instant) depends neither on the initial share of impatient respondents nor on the rate at which respondents become impatient. Reducing either does not lower the suspicion attached to any given reply; it only reshapes the distribution of reply times.

The questioner's own policy is absorbed in a similar way. A more lenient questioner, who investigates only at higher posteriors, faces proportionally higher suspicion odds. Yet the probability that any given reply is investigated, and the timing of honest replies, are unchanged; only the share of fabricated replies rises.

The invariance result has three sources that combine. First, only impatient respondents fabricate: patient ones strictly prefer to wait, so the binding indifference condition involves only the impatient type's parameters. Second, the continuous-time structure decouples the suspicion path from the fabrication rate: as long as enough impatient respondents remain to supply it, the fabrication rate adjusts to keep the impatient type indifferent, and the suspicion path inherits its shape from the indifference condition alone. Third, impatience is absorbing, so the impatient type's continuation value involves her own discount rate, the arrival rate θ of a well-considered answer, the reward, and the cost functions, but neither the initial share of impatient respondents nor the rate at which others become impatient. The substantive implication is that anti-fabrication policies that operate by reducing the supply of fabricators, for instance through detection software or bans on generative tools, cannot reduce equilibrium suspicion while enough

impatient respondents remain. They only redistribute when replies arrive.

The model sits closest to the literature on dynamic information disclosure, in which informed agents choose when to release verifiable information (Acharya, DeMarzo, and Kremer, 2011; Guttman, Kremer, and Skrzypacz, 2014; Hopenhayn and Squintani, 2011, 2016). The departure here is that the information need not exist at the time of release: the respondent may fabricate, and the strategic interaction between the timing of genuine and fake disclosure is what drives the results. The paper is also a continuous-time inspection game in the sense of Rothenstein and Zamir (2002) and Avenhaus and Canty (2005), with fabrication as the hidden wrongdoing and the questioner's suspicion as the implicit inspection intensity. The invariance result has a distant analogue in Tsebelis (1989), who observed that in static inspection games the equilibrium probability of wrongdoing is insensitive to the severity of punishment; our invariance concerns a different primitive (the supply of potential fabricators rather than the punishment) and arises through a different mechanism (the binding indifference condition is the wrongdoer's own rather than the inspector's), but shares the feature that a parameter intuition suggests should matter is silently absorbed by an equilibrium indifference condition.

The incentive to hold back a good answer is reminiscent of the ratchet effect (Weitzman, 1980; Freixas, Guesnerie, and Tirole, 1985), in which an agent deliberately underperforms so as not to reveal a high productivity that the principal would exploit in later periods, an effect that outside labor markets can mitigate (Kanemoto and MacLeod, 1992). In both settings the agent sacrifices current performance to manage an inference. The differences lie in what is inferred and why the inference is costly. Under the ratchet effect, the inference concerns a persistent type and hurts through future contracts. Here it concerns the quality of a single reply and hurts immediately, and the delay is driven not by the respondent's own future but by the presence of other types who might fabricate.

The same distinction separates the paper from reputation and career-concerns models, in which experts distort their reports or actions to appear able (Morris, 2001; Ely and Välimäki, 2003; Ottaviani and Sørensen, 2006a,b). Closest to us in that tradition are models in which timing shapes reputation. In Bonatti and Hörner (2017), the market learns about a worker's ability from whether and when a breakthrough occurs. In Tajika (2021), a reputation-concerned expert

speaks before a more accurate signal arrives in order to appear able, so speed signals quality and reputation induces haste. Here the inference runs the other way: because fabrication is fast, slowness signals quality and suspicion induces delay. Moreover, timing here signals not who the respondent is but what she did, namely whether a given answer was thought through. The mechanism therefore operates even in one-shot interactions such as surveys, interviews, and expert testimony.

A natural extension connects to a further literature. Allowing waiting to be intrinsically valuable, which we leave for future work, would relate the model to settings in which an agent delays disclosure so that an idea can mature, at the risk of being preempted (Hopenhayn and Squintani, 2011; Bobtcheff, Bolte, and Mariotti, 2017). Finally, when the cost of suspicion is read as a direct disutility from being suspected, the model is a psychological game in the sense of Geanakoplos, Pearce, and Stacchetti (1989) and Battigalli and Dufwenberg (2009), in which payoffs depend on others' beliefs; in our baseline the same costs arise from the questioner's optimal investigation decision, so the results do not hinge on belief-dependent preferences.

The paper proceeds as follows. Section 2 sets up the model. Section 3 constructs the equilibrium and establishes the three results. Section 4 discusses comparative statics in the reward and in the questioner's leniency, and the effect of deadlines. Section 5 concludes. All proofs are in Appendix A.

2. Model

2.1. Players, timing, and the arrival of an answer

A respondent (she) is asked a question by a questioner (he). Time is continuous and the horizon is infinite: $t \in [0, \infty)$. The respondent is either patient, with discount rate r^L, or impatient, with discount rate r^H, where $0 \leqslant r^L < r^H$; her type is private information. At time 0 she is impatient with probability $q \in (0, 1)$. While a patient respondent is silent and has not yet thought of a well-considered answer, she becomes impatient at Poisson rate $\lambda \geqslant 0$. Impatience is absorbing, and the respondent's type is fixed once an answer has come to mind. The switch captures a respondent who, as time passes, tires of the question or is tempted by a tool that would answer

it for her. Section 3.5 explains why the case of fixed types ($\lambda = 0$) admits no equilibrium of the kind studied here.

At each instant at which the respondent has not yet replied, a well-considered answer comes to her mind with instantaneous probability $\theta(t)\,dt > 0$, independently of r. We call θ the *discernment* and assume $\underline{\theta} := \inf_t \theta(t) > 0$, and $\bar{\theta} := \sup_t \theta(t) < \infty$. The shape of θ captures the nature of the question:

Scenario 1. *Deliberation pays off*: $\dot{\theta}(t) > 0$. A good answer becomes more likely as the respondent thinks, consults, or gathers information.

Scenario 2. *Fading concentration*: $\dot{\theta}(t) < 0$. Fatigue or loss of focus makes a good answer less likely over time.

Scenario 3. *Memoryless*: $\dot{\theta}(t) = 0$. The arrival is a stationary Poisson process.

We assume that the respondent thinks of at most one well-considered answer during the game.

At any time, the respondent may either reply or not. A reply can take one of two forms: a *well-considered* reply, available only after an answer has come to mind, or a *fabricated* reply, available at any time. The questioner cannot directly observe which form the reply takes; he observes only the time at which it is delivered. Once the respondent replies, the game ends.

2.2. Payoffs

Replying yields a gross reward $R > 0$, interpreted either as a direct payment or as relief from the cost of keeping the questioner waiting. If the respondent thinks of an answer at time t but delays replying until time $t' \geqslant t$, she incurs a flow cost $k(t' - t)$ per unit of time over the delay interval, where $k(0) \geqslant 0$ and k is strictly increasing.[1] A fabricated reply is assumed to be available at no direct cost beyond whatever inference the questioner draws from the timing.

[1] Strict monotonicity of k ensures that in equilibrium the reply time is a strictly increasing function of the time at which a well-considered answer arrives; see the proof of Lemma 2.

2.3. The cost of suspicion

After observing the reply time, the questioner forms a posterior belief $\rho(t) \in [0,1]$ that the reply delivered at time t was fabricated. We call it his *suspicion*.

The respondent's payoff depends on the questioner's suspicion through two cost functions: $x_c(\rho)$ if her reply is well considered and $x_\iota(\rho)$ if it is fabricated. The equilibrium analysis in Section 3 uses only the properties of these functions collected in Assumption 1 below. We first present our leading interpretation, in which the questioner decides whether to investigate a reply and the respondent bears the cost of an investigation; this makes the questioner an active player, and Section 4 exploits its structure. Other interpretations are discussed at the end of this subsection.

In the leading interpretation, which we call the *investigation model*, the questioner may investigate the reply, and he decides whether to do so on the basis of his suspicion, refined by a noisy signal.

Specifically, the questioner observes a binary signal $s \in \{\iota, c\}$ with $\Pr(s = \iota \mid \text{fabricated}) = \Pr(s = c \mid \text{well-considered}) = \alpha$. The precision α is drawn from a distribution H on $(1/2, 1)$ and observed by the questioner. He investigates if and only if his posterior after s is at least $w \in (0,1)$. Here w is the ratio of the cost of an investigation to the benefit of uncovering a fabrication, so a higher w describes a more lenient questioner. This rule is the questioner's best response at every information set.

An investigation costs the respondent $c_I > 0$, whatever the reply. The reply is then judged fabricated with probability $1 - \eta$ if it was fabricated and with probability $\eta' \in [0, 1-\eta)$ if it was well considered, and a reply judged fabricated incurs a penalty $c_A > 0$. The expected cost of an investigation is therefore

$$c_F := (1-\eta)c_A + c_I \quad \text{for a fabricated reply,} \qquad c_G := \eta' c_A + c_I \quad \text{for a well-considered one,}$$

with $c_F > c_G$. Even an honest respondent faces investigation and may be falsely charged. This *false-charge aversion* is what makes suspicion costly for honest respondents. Fabricating also carries a direct cost $c_D \geqslant 0$.

Let

$$\bar{w}(\rho) \coloneqq \frac{(1-\rho)\,w}{\rho\,(1-w)+(1-\rho)\,w}.$$

After $s = \iota$ the questioner investigates if and only if $\alpha \geqslant \bar{w}(\rho)$, and after $s = c$ if and only if $\alpha \leqslant 1 - \bar{w}(\rho)$. Hence the probabilities that a fabricated and a well-considered reply are investigated are

$$\begin{aligned} x_F(\rho) &= \int_{1/2}^{1-\bar{w}(\rho)} (1-\alpha)\,dH(\alpha) + \int_{\bar{w}(\rho)}^{1} \alpha\,dH(\alpha), \\ x_G(\rho) &= \int_{1/2}^{1-\bar{w}(\rho)} \alpha\,dH(\alpha) + \int_{\bar{w}(\rho)}^{1} (1-\alpha)\,dH(\alpha), \end{aligned} \tag{1}$$

where integrals over empty ranges are zero. The respondent's expected cost from suspicion is

$$x_\iota(\rho) \coloneqq c_F\,x_F(\rho) + c_D \quad \text{if she fabricates,} \qquad x_c(\rho) \coloneqq c_G\,x_G(\rho) \quad \text{if her reply is well considered.} \tag{2}$$

Lemma 1. *Suppose H has a density h that is positive and continuously differentiable on $[1/2, 1]$ with $h'(1/2) = 0$. Then x_F and x_G are twice continuously differentiable and strictly increasing on $[0, 1)$, with $x_F(0) = x_G(0) = 0$ and $x_F(1) = x_G(1) = 1$.*

The analysis in Section 3 uses only the following properties of the cost of suspicion.

Assumption 1 (Cost of suspicion)**.**

1. x_c and x_ι are twice continuously differentiable on $[0, 1)$, with $x_c' > 0$ and $x_\iota' > 0$.
2. $x_\iota(0) < R < x_\iota(1)$. Let $\widehat{\rho} \coloneqq x_\iota^{-1}(R) \in (0, 1)$.
3. $x_c \geqslant 0$ and $x_c(\widehat{\rho}) < R$.

Part 2 states that fabrication is attractive if no suspicion is attached to it and unattractive if the questioner is certain that the reply is fabricated; $\widehat{\rho}$ is the suspicion level at which a fabricated reply yields zero net payoff. Part 3 states that a well-considered reply is worth delivering even at that level.

In the investigation model, Assumption 1 holds under the assumptions of Lemma 1 together with

$$c_D < R, \qquad c_G < R < c_F + c_D.$$

Indeed, then $x_\iota(0) = c_D < R < c_F + c_D = x_\iota(1)$ and $x_c \leqslant c_G < R$.

We restrict attention to belief paths with values in $[0, \widehat{\rho}]$. On the equilibrium path this is without loss of generality: at a time t with $\rho(t) > \widehat{\rho}$, fabrication would yield a negative payoff and would not occur, so Bayes' rule would give $\rho(t) = 0$ whenever well-considered replies are delivered at t. On the compact interval $[0, \widehat{\rho}]$, both x_ι' and x_c' are bounded away from zero and infinity.

As noted above, Section 3 applies to any cost functions satisfying Assumption 1. Two other readings are natural. In the first, the respondent is directly averse to being suspected, a psychological-game formulation in the sense of Geanakoplos, Pearce, and Stacchetti (1989) and Battigalli and Dufwenberg (2009), and x_c, x_ι are primitives satisfying Assumption 1. In the second, her cost is the questioner's expected posterior after a noisy verification signal; Assumption 1 then holds whenever $R \in (0, 1)$, because the questioner's posterior is a martingale.

We study perfect Bayesian equilibrium. Strategies and beliefs are defined formally in the next section. For any function $\varphi(t)$, $\dot{\varphi}(t)$ denotes its time derivative. For a constant a, we write $\varphi \equiv a$ when $\varphi(t) = a$ for all t.

3. Equilibrium

3.1. Strategies, beliefs, and equilibrium concept

A (behavioral) strategy for the respondent specifies two objects. If a respondent of type r thinks of a well-considered answer at time t, she delivers it at time $\sigma_r(t) \geqslant t$. If she is of type r at time t and has not yet thought of an answer, she fabricates a reply with instantaneous rate $f_r(t) \in \mathbb{R}_+$, so that the probability of fabricating during $[t, t + dt)$ is $f_r(t)\, dt$. Let $\sigma = (\sigma_r)_r$ and $f = (f_r)_r$. Throughout we focus on strategies in which σ_r is differentiable with $\dot{\sigma}_r(t) > 0$, so that a respondent who thinks of an answer earlier replies earlier.

Given such a strategy, Bayesian consistency pins down the questioner's suspicion $\rho(t)$ that a

reply delivered at time t is fabricated. Let $N_H(t)$ and $N_L(t)$ denote the probabilities that at time t the respondent is silent, has not yet thought of an answer, and is impatient or patient, respectively, and write $\Theta(t) \coloneqq \int_0^t \theta(s)\,ds$. These probabilities evolve according to

$$\dot{N}_L(t) = -\big(\theta(t) + \lambda + f_{r^L}(t)\big)N_L(t), \qquad \dot{N}_H(t) = \lambda N_L(t) - \big(\theta(t) + f_{r^H}(t)\big)N_H(t), \tag{3}$$

with $N_L(0) = 1 - q$ and $N_H(0) = q$. A respondent of type r who thinks of an answer at s replies at $\sigma_r(s)$, so the density of well-considered replies at t is[2]

$$\begin{aligned} &G(t)\,dt = \theta(s_H)\,N_H(s_H)\,ds_H + \theta(s_L)\,N_L(s_L)\,ds_L, \qquad s_H = \sigma_{r^H}^{-1}(t),\ s_L = \sigma_{r^L}^{-1}(t), \\ &\Rightarrow G(t) \coloneqq \frac{\theta(s_H)\,N_H(s_H)}{\dot{\sigma}_{r^H}(s_H)} + \frac{\theta(s_L)\,N_L(s_L)}{\dot{\sigma}_{r^L}(s_L)}, \end{aligned}$$

where a term is zero when $t < \sigma_r(0)$. The density of fabricated replies is $F(t) \coloneqq f_{r^H}(t)N_H(t) + f_{r^L}(t)N_L(t)$. By Bayes' rule,

$$\rho(t;\sigma,f) = \begin{cases} \dfrac{F(t)}{F(t)+G(t)} & \text{if } F(t)+G(t) > 0, \\ \text{arbitrary} & \text{otherwise.} \end{cases} \tag{4}$$

Now fix a belief path ρ and consider the respondent's problem. If she has thought of an answer at time t, her type is fixed and her continuation value when choosing the reply time $\sigma(t) \geqslant t$ is

$$V_r^A(t;\rho) = \max_{\sigma(t)\geqslant t} e^{-r(\sigma(t)-t)}\,[R - x_c(\rho(\sigma(t)))] - \int_t^{\sigma(t)} e^{-r(\tau-t)}\,k(\tau-t)\,d\tau. \tag{5}$$

If she is impatient and has not yet thought of an answer, her value $V_{r^H}^W(t;\rho)$ satisfies the Bellman equation

$$\begin{aligned} V_{r^H}^W(t;\rho) = \max_{f(t)} \Big\{ &f(t)\,dt\,(R - x_\iota(\rho(t))) + \theta(t)\,dt\,V_{r^H}^A(t;\rho) \\ &+ e^{-r^H dt}(1 - \theta(t)\,dt - f(t)\,dt)\,V_{r^H}^W(t+dt;\rho)\Big\}. \end{aligned} \tag{6}$$

[2]The density of the times at which answers come to mind is converted into a density of reply times. Answers that come to mind in $[s, s+ds]$ are delivered in $[t, t+dt]$ with $dt = \dot{\sigma}_r(s)\,ds$, so the density is divided by $\dot{\sigma}_r(s)$. Without delay, $\dot{\sigma}_r = 1$.

If she is patient, she may in addition become impatient:

$$V^W_{rL}(t;\rho) = \max_{f(t)} \Big\{ f(t)\,dt\,(R - x_\iota(\rho(t))) + \theta(t)\,dt\,V^A_{rL}(t;\rho) + \lambda\,dt\,V^W_{rH}(t;\rho) \\ + e^{-r^L dt}(1 - \theta(t)\,dt - \lambda\,dt - f(t)\,dt)\,V^W_{rL}(t+dt;\rho) \Big\}. \tag{7}$$

Both value functions are strictly positive: $V^A_r > 0$ because $\sigma(t) = t$ delivers $R - x_c(\rho(t)) > 0$, and $V^W_r > 0$ because never fabricating delivers at least the discounted value of eventually thinking of an answer.

Definition 1. A triple (σ, f, ρ) is a *perfect Bayesian equilibrium* (PBE) if, for each r and t, $\sigma_r(t)$ attains the maximum in (5), $f_r(t)$ attains the maximum in (6) or (7), and ρ satisfies (4).

We restrict attention throughout to PBEs in which ρ is continuously differentiable.

3.2. Existence and the invariance result

We begin with three preliminary results. The first establishes monotonicity of σ_r, which lets us invert it.

Lemma 2. *In any PBE with differentiable ρ, σ_r is strictly increasing.*

Taking $dt \to 0$ in (6) yields a clean ODE for the impatient type's value. Because impatience is absorbing, this ODE is unaffected by λ.

Lemma 3. *In any PBE,*

$$(r^H + \theta(t))\,V^W_{rH}(t;\rho) - \dot{V}^W_{rH}(t;\rho) = \theta(t)\,V^A_{rH}(t;\rho), \tag{8}$$

and $R - x_\iota(\rho(t)) \leqslant V^W_{rH}(t;\rho)$ for every t, with equality whenever $f_{rH}(t) > 0$.

Boundedness of V^W_{rH} selects the constant of integration and yields

$$V^W_{rH}(t;\rho) = e^{\Delta(t)} \int_t^\infty e^{-\Delta(t')}\,\theta(t')\,V^A_{rH}(t';\rho)\,dt', \qquad \Delta(t) := r^H t + \Theta(t). \tag{9}$$

Patient respondents wait both because they discount less and because they may later become impatient and fabricate then. Either way, fabricating now is never attractive to them.

Lemma 4. *In any PBE, $f_{r^L} \equiv 0$: patient respondents never fabricate.*

Hence only impatient respondents fabricate. If they fabricate with positive intensity at t, Lemma 3 gives

$$R - x_\iota(\rho(t)) = V^W_{r^H}(t;\rho) \iff \rho(t) = x_\iota^{-1}\Big(R - V^W_{r^H}(t;\rho)\Big). \tag{10}$$

Equation (10) defines the self-mapping

$$\Psi[\rho](t) := x_\iota^{-1}\left(R - e^{\Delta(t)}\int_t^\infty e^{-\Delta(t')}\,\theta(t')\,V^A_{r^H}(t';\rho)\,dt'\right). \tag{11}$$

Hence, in any PBE in which the impatient type fabricates with positive intensity at every instant, the suspicion path is a fixed point of Ψ. Because Ψ involves neither q, nor λ, nor r^L, neither does the set of such suspicion paths; this is the source of the invariance in Theorem 1 below. Let $\underline{x}'_\iota := \min_{p\in[0,\widehat{\rho}]} x'_\iota(p) > 0$, $\bar{x}'_c := \max_{p\in[0,\widehat{\rho}]} x'_c(p) < \infty$, and $\beta_1 := \bar{\theta}R/\underline{x}'_\iota$, and define

$$\underline{\rho} := x_\iota^{-1}\Big(\frac{r^H}{r^H+\bar{\theta}}\,R\Big), \qquad \bar{\rho} := x_\iota^{-1}\Big(R - \frac{\underline{\theta}}{r^H+\underline{\theta}}\,(R - x_c(\widehat{\rho}))\Big),$$

$$\mathcal{K} := \big\{\rho : \mathbb{R}_+ \to [\underline{\rho},\bar{\rho}] \;\big|\; |\rho(t) - \rho(t')| \leqslant \beta_1\,|t-t'| \text{ for all } t,t'\big\}.$$

Lemma 5. *If $r^H > \bar{\theta}\,x_\iota(0)/(R - x_\iota(0))$, then $0 < \underline{\rho} \leqslant \bar{\rho} < \widehat{\rho}$, Ψ maps every continuous belief path into $\mathcal{K}$, and Ψ has a continuously differentiable fixed point $\rho^* \in \mathcal{K}$.*

The fixed point ρ^* depends only on r^H, θ, R, k, x_c, and x_ι. In particular, it depends neither on q, nor on λ, nor on r^L. Since $|\dot{\rho}^*| \leqslant \beta_1$, suspicion can fall only at a bounded rate. The only gain from delaying a well-considered reply is this fall in suspicion, which lowers the cost $x_c(\rho)$ at rate at most $\bar{x}'_c\beta_1$, whereas delay costs a sufficiently impatient respondent more than that through discounting alone. Such a respondent therefore never delays a well-considered reply.

Lemma 6. *Suppose $\rho \in \mathcal{K}$ is continuously differentiable and*

$$r^H \left(R - x_c(\widehat{\rho})\right) > \bar{x}_c' \, \beta_1. \tag{12}$$

Then $\sigma_{r^H}(t) = t$ for all t.

Condition (12) is stated entirely in terms of primitives: $\beta_1 = \bar{\theta} R / \underline{x}_\iota'$, $\widehat{\rho}$, and $\bar{x}_c'$ depend only on θ, R, x_c, and x_ι, and not on r^H, q, or λ. Since its left-hand side grows linearly in r^H, it holds whenever the impatient type is sufficiently impatient, as does the condition $r^H > \bar{\theta}\, x_\iota(0)/(R - x_\iota(0))$ of Lemma 5. No equilibrium object enters either condition.

Bayes' rule involves the density of the patient type's delayed replies, which requires her reply-time function to be well behaved. We impose the following regularity condition.

Assumption 2 (Regularity)**.** $\theta \in C^1$ with $\sup_t |\dot{\theta}(t)| < \infty$, and $k \in C^1$ with $\inf_{\tau \geqslant 0} k'(\tau) > M$, where M is the constant defined in (25) in the proof of Lemma 7.

The constant M depends only on primitives: r^L, r^H, R, $\bar{\theta}$, $\sup_t |\dot{\theta}(t)|$, and bounds on the first and second derivatives of x_c and x_ι on $[0, \widehat{\rho}]$. It bounds how fast the patient type's marginal gain from waiting, which comes from the falling suspicion, can change over time. The assumption says that the marginal cost of delay rises faster than this gain can change, so that each patient respondent's choice of delay is single-peaked.

Lemma 7. *Suppose (12) and Assumption 2 hold. Then ρ^* is twice continuously differentiable with bounded second derivative. For each t, problem (5) for $r = r^L$ has a unique solution $\sigma^*_{r^L}(t)$. The map $\sigma^*_{r^L}$ is continuous and strictly increasing, and it is continuously differentiable with $\dot{\sigma}^*_{r^L}(t) > 1/2$ wherever $\sigma^*_{r^L}(t) > t$.*

Hence the density of the patient type's well-considered replies is well defined almost everywhere.

Let σ^* denote the reply times that solve (5) given ρ^*, so that $\sigma^*_{r^H}(t) = t$ under (12), and define

$$G^*_L(t) := \begin{cases} \dfrac{\theta(s)\, N_L(s)}{\dot{\sigma}^*_{r^L}(s)}, & s = (\sigma^*_{r^L})^{-1}(t), \quad t \geqslant \sigma^*_{r^L}(0), \\ 0 & \text{otherwise}, \end{cases} \tag{13}$$

where $N_L(t) = (1-q)e^{-\Theta(t)-\lambda t}$ by Lemma 4. Here G_L^* is the density of the patient type's well-considered replies under σ^*. If the impatient type fabricates at every instant so as to keep $\rho = \rho^*$, Bayes' rule (4) requires

$$f_{r^H} N_H = \frac{\rho^*}{1-\rho^*} G = \frac{\rho^*}{1-\rho^*} (\theta N_H + G_L^*) :$$

each well-considered reply must be accompanied by $\rho^*/(1-\rho^*)$ units of fabrication. Substituting this into (3) yields a linear ODE for N_H, whose solution is

$$N_H^*(t) = e^{-C(t)} \left[q + \int_0^t e^{C(s)} \left(\lambda N_L(s) - \frac{\rho^*(s)}{1-\rho^*(s)} G_L^*(s) \right) ds \right], \qquad C(t) := \int_0^t \frac{\theta(s)}{1-\rho^*(s)} ds. \tag{14}$$

The function N_H^* is the probability that a silent impatient respondent without an answer remains available to fabricate. We say that fabricators are *abundant* if $N_H^*(t) > 0$ for all $t > 0$. Unlike N_L, which declines at the bounded per-capita rate $\theta+\lambda$ and therefore stays positive, N_H^* can reach zero in finite time: Bayes' rule requires fabrication in proportion to the patient type's well-considered replies, and this requirement does not shrink as the pool of impatient respondents runs down.

Theorem 1. *Suppose $r^H > \bar{\theta}\, x_\iota(0)/(R - x_\iota(0))$, (12) and Assumption 2 hold, and fabricators are abundant. Then there exists a PBE in which the suspicion path is ρ^*, only impatient respondents fabricate, and they do so with intensity*

$$f_{r^H}(t) = \frac{\rho^*(t)}{1-\rho^*(t)} \cdot \frac{\theta(t) N_H^*(t) + G_L^*(t)}{N_H^*(t)}. \tag{15}$$

The suspicion path ρ^* is independent of the initial share q of impatient respondents, of the rate λ at which respondents become impatient, and of the patient type's discount rate r^L.

The invariance is structurally the same indifference-absorbs-primitives phenomenon familiar from static inspection games (Tsebelis, 1989): the suspicion path is pinned down by the impatient type's indifference condition (10), which involves only r^H, θ, R, k, and the cost functions. The supply of fabricators, governed by q and λ, shifts the *rate* at which each impatient respondent fabricates, through (15), and hence the timing distribution of replies, but not the suspicion

attached to any given reply time.

Abundance is defined through equilibrium objects, but it is implied by a condition on primitives. Let

$$\bar{d} \coloneqq \frac{r^L R + \bar{x}_c' \beta_1}{\inf_{\tau \geqslant 0} k'(\tau)}.$$

As shown in the proof below, no patient respondent delays a well-considered reply by more than $\bar{d}$. Recall also that $\rho^* \leqslant \bar{\rho}$, the upper bound in Lemma 5.

Corollary 1. *Under the assumptions of Theorem 1 other than abundance, fabricators are abundant if*

$$\lambda \geqslant \frac{\bar{\rho}}{1-\bar{\rho}}\,\bar{\theta}\,\exp\!\Big(\frac{\bar{\theta}\,\bar{d}}{1-\bar{\rho}}\Big). \tag{16}$$

Condition (16) says that the flow of newly impatient respondents must be large enough to supply the fabrication that Bayes' rule requires to accompany the patient type's well-considered replies. The exponential factor accounts for the delay of those replies: a reply delivered at t comes from an answer that arrived up to $\bar{d}$ earlier, when more patient respondents were still silent.

Remark 1 (More types). The argument extends to finitely many discount rates $r_1 < \cdots < r_n$ when a respondent without an answer can only switch to a higher discount rate. The highest type is absorbing, so its indifference condition again defines Ψ, and ρ^* is pinned down by that type's parameters alone. The proof of Lemma 4 extends by induction from r_{n-1} down to r_1: the difference between type r_i's value from never fabricating and $V^W_{r_n}$ solves an equation like the one in that proof, with additional nonnegative terms $\lambda_{ij}(V_{r_j} - V^W_{r_n})$ for the switches to $j > i$, so it is positive. Hence only the highest type fabricates, provided an analogous abundance condition holds for it, with λN_L replaced by the total flow into the highest type.

3.3. The shape of the suspicion path

Existence does not by itself tell us what ρ^* looks like over time. The next result ties the monotonicity of ρ^* directly to the shape of the discernment.

Proposition 1. *Suppose* $r^H > \bar{\theta}\, x_\iota(0)/(R - x_\iota(0))$.

1. *If* $\dot{\theta}(t) > 0$ *for all* t, Ψ *has a fixed point* ρ^* *that is strictly decreasing.*

2. *If* $\dot{\theta}(t) < 0$ *for all* t, Ψ *has a fixed point* ρ^* *that is strictly increasing.*

3. *If* $\dot{\theta}(t) = 0$ *for all* t, Ψ *has a fixed point* ρ^* *that is constant.*

The intuition is direct. When discernment is improving, a late reply is a priori more likely to reflect a genuinely arrived answer, so suspicion falls over time. When discernment is fading, the opposite holds. In the stationary Poisson case, the passage of time carries no information and suspicion is flat.

In each monotone equilibrium, the limit $\rho^*(\infty)$ is pinned down by a simple indifference equation.

Lemma 8. *Suppose* θ *is monotone, and let* ρ^* *be a monotone fixed point of* Ψ. *Then* $\rho^*(t) \to \rho^*(\infty)$, *where* $\rho^*(\infty)$ *solves*

$$\rho^*(\infty) = x_\iota^{-1}\left(R - \frac{\theta(\infty)}{\theta(\infty) + r^H}\left(R - x_c(\rho^*(\infty))\right)\right), \tag{17}$$

equivalently

$$x_\iota(\rho^*(\infty)) - \frac{\theta(\infty)}{\theta(\infty) + r^H} x_c(\rho^*(\infty)) = \frac{r^H}{\theta(\infty) + r^H} R. \tag{18}$$

Equation (18) has a transparent interpretation. The left-hand side is the net expected suspicion cost of fabricating rather than waiting: $x_\iota(\rho^*(\infty))$ is the cost paid immediately by a fabricator, and $\frac{\theta(\infty)}{\theta(\infty)+r^H} x_c(\rho^*(\infty))$ is the discounted expected cost she would have paid after eventually arriving at a well-considered reply. The right-hand side is the benefit of fabricating: the fraction $\frac{r^H}{\theta(\infty)+r^H}$ of the reward R that the impatient type saves by not waiting.

Because $\Psi[\rho](t)$ depends only on $\rho(t')$ for $t' \geqslant t$, the suspicion path is determined by forward-looking indifference. A simple sufficient condition for uniqueness is that Ψ be a contraction.

Lemma 9. *Let* $\kappa := \dfrac{\bar{x}_c'\,\bar{\theta}}{\underline{x}_\iota'\,(r^H + \bar{\theta})}$. *Then* $\sup_t |\Psi[\rho](t) - \Psi[\omega](t)| \leqslant \kappa \sup_t |\rho(t) - \omega(t)|$ *for all continuous belief paths* ρ *and* ω. *Hence, if* $\kappa < 1$, Ψ *has a unique fixed point.*

Corollary 2. *Under the assumptions of Theorem 1, suppose $\kappa < 1$. Then ρ^* is the unique fixed point of Ψ, so the suspicion path is the same in every PBE in which the impatient type fabricates at every instant. Consequently, by Proposition 1, ρ^* is strictly decreasing if $\dot{\theta} > 0$, strictly increasing if $\dot{\theta} < 0$, and constant if $\dot{\theta} \equiv 0$.*

3.4. Intentional delay

We can now characterize when fear of suspicion induces the respondent to hold back a reply she already has.

Definition 2. A PBE (σ, f, ρ) exhibits *no intentional delay* if $\sigma_r(t) = t$ for all t and r.

Given a belief path ρ, let

$$\Phi_r(t) := -r\big(R - x_c(\rho(t))\big) - x_c'(\rho(t))\,\dot{\rho}(t). \tag{19}$$

This is the marginal gain to a type-r respondent from delaying a well-considered reply at t, before the cost of delay: the fall in the cost of suspicion minus the loss from discounting. Differentiating (5), the derivative of the objective with respect to the delay $\tau = \sigma(t) - t$ is $e^{-r\tau}\big[\Phi_r(t+\tau) - k(\tau)\big]$. This yields the following first-order characterization.

Lemma 10. *A PBE (σ, f, ρ) exhibits no intentional delay if and only if for each t and each r,*

$$\Phi_r(t) \leqslant k(0). \tag{20}$$

When discernment is constant or decreasing, Corollary 2 gives $\dot{\rho}^* \geqslant 0$, so $\Phi_r \leqslant 0 \leqslant k(0)$ for every r: no intentional delay arises.

Corollary 3. *Suppose the assumptions of Corollary 2 hold, and either $\dot{\theta}(t) < 0$ for all t or $\dot{\theta} \equiv 0$. Then the PBE exhibits no intentional delay.*

When discernment is improving, by contrast, $\dot{\rho}^* < 0$, and a sufficiently patient respondent finds it worthwhile to wait in order to benefit from a lower suspicion in the future.

Theorem 2 (Strategic delay)**.** *Suppose* $\dot{\theta}(t) > 0$ *for all* t*,* $k(0) = 0$*, and the assumptions of Theorem 1 hold. If* r^L *is sufficiently small, then in the PBE of Theorem 1 with a nonincreasing* ρ^**, patient respondents intentionally delay over some interval of time.*

A nonincreasing fixed point ρ^* exists by Proposition 1, and it is the only one if $\kappa < 1$. Abundance depends on r^L through the patient type's replies, but the primitive condition (16) only becomes easier to satisfy as r^L falls, because $\bar{d}$ is increasing in r^L.

The condition $k(0) = 0$ can be weakened to $k(0) < -x_c'(\rho^*(t))\,\dot{\rho}^*(t)$ for some t. Some such condition is needed: if $k(0) > \bar{x}_c'\beta_1$, the cost of delaying exceeds any gain from lower future suspicion, and no one ever delays.

Because ρ^* is independent of q and λ, so is the equilibrium delay function $\sigma_r(t) - t$, as long as fabricators are abundant. Reducing the supply of potential fabricators does not shorten the delay of honest respondents.

3.5. Abundant and scarce fabricators

The assumption that patient respondents may become impatient is not cosmetic. If types were fixed ($\lambda = 0$), the pool of potential fabricators would be exhausted in finite time.

Proposition 2. *Suppose* $\lambda = 0$*,* $r^H > \bar{\theta}\,x_\iota(0)/(R - x_\iota(0))$*, and* (12) *and Assumption 2 hold. Then there is no PBE in which the impatient type fabricates with positive intensity at every instant.*

The logic is simple. Patient respondents never fabricate, yet they keep delivering well-considered replies, and Bayes' rule requires each such reply to be accompanied by a proportional amount of fabrication. This creates an asymmetry between the two pools. The pool of patient respondents declines at the bounded per-capita rate θ, so it shrinks exponentially but never empties. The pool of impatient respondents must in addition supply fabrication in proportion to the patient type's replies, and this requirement does not shrink as the impatient pool runs down. With fixed types, the impatient pool is therefore exhausted in finite time. When patient respondents become impatient at a positive rate, the pool is replenished, and Corollary 1 gives conditions under which it never runs dry.

When fabricators are scarce, so that N_H^* would reach zero at some finite time, ρ^* cannot be sustained. On any interval on which no silent impatient respondent without an answer remains, fabrication can come only from respondents who have just become impatient, at most at the flow $\lambda N_L(t)$. Bayes' rule then imposes the supply bound

$$\rho(t) \leqslant \frac{\lambda N_L(t)}{\lambda N_L(t) + G(t)},$$

which depends on q and λ. Wherever this bound binds, suspicion lies strictly below the level dictated by the impatient type's indifference, and the impatient type strictly prefers to fabricate as soon as she becomes impatient. This suggests that suspicion is pinned down by the fabricators' indifference when they are abundant and by their supply when they are scarce. A full characterization of the scarce regime, which requires handling atoms of fabrication at switching times, is left for future work.

The two regimes give the invariance result a sharp policy reading. Measures that reduce the supply of fabricators leave suspicion and the delay of honest respondents unchanged as long as fabricators remain abundant. Examples are screening out impatient respondents (q) and making it harder to become one (λ), for instance by detecting or prohibiting generative tools. By Corollary 1, abundance holds whenever λ satisfies the primitive condition (16). Such measures can lower suspicion only once they push λ below the level at which abundance fails, which lies below the right-hand side of (16). From that point on, the supply of fabricators, rather than their indifference, governs inference.

4. Discussion

4.1. Comparative statics: the reward and the questioner's leniency

A natural policy lever is the reward R. A larger reward makes fabrication more tempting, which raises equilibrium suspicion, which in turn erodes the net payoff $R - x_c(\rho)$ of a well-considered reply. To see which effect dominates, consider the stationary case $\theta(t) \equiv \theta$. Under (12) the

impatient type replies at once, so $V^A_{rH} = R - x_c(\rho)$, and the constant suspicion solves

$$x_\iota(\rho) = R - Q\,(R - x_c(\rho)), \qquad Q \coloneqq \frac{\theta}{\theta + r^H}.$$

Differentiating,

$$\frac{\partial \rho}{\partial R} = \frac{1 - Q}{x'_\iota(\rho) - Q\,x'_c(\rho)}, \qquad \frac{\partial (R - x_c(\rho))}{\partial R} = \frac{x'_\iota(\rho) - x'_c(\rho)}{x'_\iota(\rho) - Q\,x'_c(\rho)}.$$

The denominators are positive when $\kappa < 1$ (Lemma 9). A higher reward therefore always raises suspicion, and it lowers the net payoff of a well-considered reply if and only if $x'_c(\rho) > x'_\iota(\rho)$, that is, if an increase in suspicion hurts honest respondents more than fabricators.

In the investigation model of Section 2.3, the ratio of the two marginal costs takes a simple form.

Lemma 11. *In the investigation model of Section 2.3, under the assumptions of Lemma 1,*

$$\frac{x'_G(\rho)}{x'_F(\rho)} = \frac{\rho}{1 - \rho} \cdot \frac{1 - w}{w}.$$

Hence $x'_c / x'_\iota = (c_G / c_F) \cdot \frac{\rho}{1-\rho} \cdot \frac{1-w}{w}$, so a higher reward lowers the net payoff of a well-considered reply exactly when

$$\frac{\rho}{1 - \rho} > \frac{c_F}{c_G} \cdot \frac{w}{1 - w},$$

that is, when suspicion is high relative to the questioner's investigation threshold. At such levels a further rise in suspicion mostly adds investigations of honest replies.

When this happens, rewarding replies backfires: honest respondents are worse off. Moreover, the cost of postponing a well-considered reply in (20) is proportional to $R - x_c(\rho)$, so their incentive to reply promptly weakens as well. When $\dot\theta > 0$, so that suspicion falls over time, this suggests that higher rewards may lengthen intentional delay. Confirming this requires comparative statics of the nonstationary path ρ^*, which we leave aside.

The questioner's leniency. In the investigation model (see Section 2.3), suspicion and the questioner's threshold enter the respondent's payoffs only through

$$\bar{w}(\rho; w) = \frac{1}{1 + \frac{\rho}{1-\rho} \cdot \frac{1-w}{w}},$$

the precision a signal must have to trigger an investigation. We call $\bar{w}$ the questioner's *effective standard*. The investigation probabilities x_F and x_G, and hence the equilibrium conditions of Section 3, depend on ρ and w only through $\bar{w}$. Equilibrium therefore pins down a path of effective standards, and a change in w is absorbed by a change in suspicion that leaves this path unchanged.

Proposition 3. *Suppose (σ^*, f, ρ^*) is a PBE as in Theorem 1 under threshold w, and let $\bar{w}^*(t) := \bar{w}(\rho^*(t); w)$. For $w' \in (0,1)$, define $\rho'(t)$ by $\bar{w}(\rho'(t); w') = \bar{w}^*(t)$, that is,*

$$\frac{\rho'(t)}{1-\rho'(t)} = \frac{w'/(1-w')}{w/(1-w)} \cdot \frac{\rho^*(t)}{1-\rho^*(t)}.$$

If fabricators are abundant under w', that is, if (14) stays positive with ρ^* replaced by ρ', then there is a PBE under w' with suspicion path ρ' and reply times σ^*, in which every reply faces the same investigation probabilities as under w. If Ψ has a unique fixed point under w', its suspicion path is ρ'.

A more lenient questioner ($w' > w$) thus faces a higher share of fabricated replies at every instant. The extra fabrication exactly offsets his leniency: the probability that any given reply is investigated is unchanged, and so are the honest respondents' exposure to false charges and their delay. As with the invariance result, an equilibrium indifference condition absorbs a change in a primitive that one might expect to matter. Because more fabrication is needed, however, abundance becomes harder to satisfy as the questioner becomes more lenient.

4.2. Deadline

Finally, consider the effect of a hard deadline D: the respondent earns R if she replies by time D and 0 otherwise. The value function after thinking of an answer becomes

$$V_r^{A,D}(t;\rho) = \max_{\sigma(t)\in[t,D]} e^{-r(\sigma(t)-t)}[R - x_c(\rho(\sigma(t)))] - \int_t^{\sigma(t)} e^{-r(\tau-t)}k(\tau-t)\,d\tau,$$

and the same argument as in Lemma 3 yields, for $r = r^H$,

$$V_r^{W,D}(t;\rho) = e^{rt+\int_0^t \theta(\tilde{t})d\tilde{t}} \int_t^D e^{-(rt'+\int_0^{t'} \theta(\tilde{t})d\tilde{t})}\,\theta(t')\,V_r^{A,D}(t';\rho)\,dt',$$

using the terminal condition $V_r^{W,D}(D;\rho) = 0$. As before, the equilibrium suspicion satisfies $\rho(t) = x_\iota^{-1}(R - V_{r^H}^{W,D}(t;\rho))$. Continuity at the deadline forces $\rho(D) = x_\iota^{-1}(R)$, and since $V_{r^H}^{W,D} \geqslant 0$, we have $\rho(D) \geqslant \rho(t)$ for all $t < D$.

Because waiting has no intrinsic benefit, the deadline case inherits the properties of the baseline, provided fabricators are abundant on $[0, D]$, that is, the analogue of (14) on $[0, D]$ stays positive. Under this condition the suspicion path is pinned down by the impatient type's indifference, so it is independent of q and λ, and patient respondents still never fabricate before D: the proof of Lemma 4 goes through with the terminal condition $V_r^{W,D}(D) = 0$. Because the horizon is finite, abundance can now hold even with fixed types ($\lambda = 0$), provided q is large enough that the initial pool of impatient respondents is not exhausted before D. Near the deadline, however, suspicion approaches $x_\iota^{-1}(R)$, so more fabrication is required there and abundance becomes harder to satisfy.

5. Conclusion

Intentional delay and fabricated replies look, at first glance, like independent frustrations of the questioner. This paper has shown that they are tightly linked: when a well-considered answer arrives at a rate that rises over time, fear of being mistaken for a fabricator induces even a respondent who has already thought of an answer to hold it back. Only impatient respondents fabricate, and as long as they are abundant, the equilibrium suspicion attached to any given reply

is independent of how many of them there are and of how quickly respondents become impatient. Suspicion, and the delay it induces, can fall only once fabricators become scarce.

The invariance has a distant cousin in Tsebelis's (1989) result for static inspection games, where the equilibrium probability of wrongdoing is insensitive to the severity of punishment. The two results share the broader feature that a parameter intuition suggests should matter is absorbed by an equilibrium indifference condition, although the parameter, the binding condition, and the equilibrium object differ in the two settings. At the same time, the invariance is not absolute: it rests on fabricators being abundant. Several questions remain open. First, when fabricators are scarce, suspicion is bounded by their supply, and a full characterization of equilibrium in that regime requires handling atoms of fabrication at the times respondents become impatient. Second, we have assumed that waiting is purely costly. If waiting has intrinsic value, for instance because an answer can be polished, a deadline may cause well-considered replies to cluster at the deadline. Bayesian consistency would then require an atom of fabrication there, which may in turn draw patient respondents into fabrication and break invariance. Third, the comparative statics of the nonstationary suspicion path, for instance with respect to the reward, remain to be worked out. More broadly, understanding when indifference conditions absorb shifts in primitives, and when they fail to, is a natural direction for further work.

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A. Proofs

Proof of Lemma 1. Note that $\bar{w}(\rho) = 1/(1+\ell)$ with $\ell \coloneqq \frac{\rho}{1-\rho} \cdot \frac{1-w}{w}$, so $\bar{w}$ is smooth and strictly decreasing on $[0,1)$, with $\bar{w}(0) = 1$ and $\bar{w}(\rho) \to 0$ as $\rho \to 1$. In particular $x_F(0) = x_G(0) = 0$ and $x_F(1) = x_G(1) = 1$.

Let $S(u) \coloneqq h(\max\{u, 1-u\})$. Differentiating (1) under the integral sign gives

$$x_F'(\rho) = -\bar{w}(\rho)\,\bar{w}'(\rho)\,S(\bar{w}(\rho)), \qquad x_G'(\rho) = -(1-\bar{w}(\rho))\,\bar{w}'(\rho)\,S(\bar{w}(\rho)).$$

Only one of the two integrals moves at a time: the first when $\bar{w} < 1/2$ and the second when $\bar{w} > 1/2$. Since $h > 0$ and $\bar{w}' < 0$, both derivatives are positive on $[0,1)$.

The function S is continuous. It is continuously differentiable because its one-sided derivatives at $u = 1/2$ are $h'(1/2)$ and $-h'(1/2)$, which coincide when $h'(1/2) = 0$. Hence x_F and x_G are C^2. □

Proof of Lemma 2. We drop the subscript r for notational simplicity. We first show that σ is nondecreasing, then that it is strictly increasing.

Suppose by contradiction that $\sigma(t) > \sigma(t')$ for some $t < t'$. For a respondent who thinks of an answer at time t, the incentive to prefer $\sigma(t)$ over $\sigma(t')$ gives

$$\begin{aligned} e^{-r(\sigma(t')-t)}[R - x_c(\rho(\sigma(t')))] - \int_t^{\sigma(t')} e^{-r(\tau-t)} k(\tau - t)\, d\tau \\ \leqslant e^{-r(\sigma(t)-t)}[R - x_c(\rho(\sigma(t)))] - \int_t^{\sigma(t)} e^{-r(\tau-t)} k(\tau - t)\, d\tau. \end{aligned} \tag{21}$$

For a respondent who thinks of an answer at time t', the reverse preference gives

$$\begin{aligned} e^{-r(\sigma(t')-t')}[R - x_c(\rho(\sigma(t')))] - \int_{t'}^{\sigma(t')} e^{-r(\tau-t')} k(\tau - t')\, d\tau \\ \geqslant e^{-r(\sigma(t)-t')}[R - x_c(\rho(\sigma(t)))] - \int_{t'}^{\sigma(t)} e^{-r(\tau-t')} k(\tau - t')\, d\tau. \end{aligned} \tag{22}$$

Combining (21) and (22) yields

$$\int_{\sigma(t')}^{\sigma(t)} e^{-r\tau} k(\tau - t')\, d\tau \geqslant \int_{\sigma(t')}^{\sigma(t)} e^{-r\tau} k(\tau - t)\, d\tau,$$

contradicting the strict monotonicity of k (since $t < t'$).

Now suppose ρ is continuously differentiable and that $\sigma(t) = \sigma(t') = t^*$ for some $t > t'$. Consider first the case $t^* > t$. Differentiability of ρ yields the first-order conditions

$$\begin{aligned} -r(R - x_c(\rho(t^*))) - x_c'(\rho(t^*))\dot{\rho}(t^*) - k(t^* - t) = 0, \\ -r(R - x_c(\rho(t^*))) - x_c'(\rho(t^*))\dot{\rho}(t^*) - k(t^* - t') = 0, \end{aligned}$$

which contradict strict monotonicity of k. Consider next the case $\sigma(t) = t$. If $\sigma(t') = t$ for some

$t' < t$, then

$$-r(R - x_c(\rho(t))) - x_c'(\rho(t))\dot{\rho}(t) - k(0) \leqslant 0,$$
$$-r(R - x_c(\rho(t))) - x_c'(\rho(t))\dot{\rho}(t) - k(t - t') = 0,$$

implying $k(0) \geqslant k(t - t')$, again contradicting strict monotonicity of k. □

Proof of Lemma 3. Taking $dt \to 0$ in (6) yields

$$(r^H + \theta(t))\, V_{rH}^W(t;\rho) - \dot{V}_{rH}^W(t;\rho) = \max_{f(t)} \left\{ f(t)\,(R - x_\iota(\rho(t)) - V_{rH}^W(t;\rho)) + \theta(t)\, V_{rH}^A(t;\rho) \right\}. \tag{23}$$

Claim 1. *In any PBE, $f_{rH}(t) < \infty$ for each t.*

Proof. Suppose by contradiction that $f_{rH}(t) = \infty$ for some t. Then $\rho(t) = 1$ by (4), so $R - x_\iota(\rho(t)) < 0 < V_{rH}^W$. This makes $f(t) = 0$ the unique maximizer in (23), a contradiction. □

If $R - x_\iota(\rho(t)) > V_{rH}^W(t;\rho)$, then $f(t) = \infty$ would be optimal, contradicting Claim 1. Hence $R - x_\iota(\rho(t)) \leqslant V_{rH}^W(t;\rho)$, with equality whenever $f_{rH}(t) > 0$. In either case the maximum equals $\theta(t) V_{rH}^A(t;\rho)$, which yields (8). □

Proof of Lemma 4. We first show that $V_r^A(t;\rho)$ is nonincreasing in r. By the envelope theorem and $\sigma_r(t) \geqslant t$,

$$\begin{aligned}\frac{\partial V_r^A(t;\rho)}{\partial r} &= -(\sigma_r(t) - t)\, e^{-r(\sigma_r(t)-t)}\,[R - x_c(\rho(\sigma_r(t)))] + \int_t^{\sigma_r(t)} (\tau - t)\, e^{-r(\tau-t)} k(\tau - t)\, d\tau \\ &\leqslant -(\sigma_r(t) - t)\, V_r^A(t;\rho) \leqslant 0.\end{aligned}$$

Let $\widehat{V}_L(t)$ be the value of a patient respondent who never fabricates while patient and behaves optimally after becoming impatient. Taking $dt \to 0$ in (7) with $f \equiv 0$ gives

$$(r^L + \theta + \lambda)\widehat{V}_L - \dot{\widehat{V}}_L = \theta V_{rL}^A + \lambda V_{rH}^W.$$

Subtracting (8),

$$(r^L+\theta+\lambda)\big(\widehat{V}_L - V^W_{r^H}\big) - \big(\dot{\widehat{V}}_L - \dot{V}^W_{r^H}\big) = \theta\,(V^A_{r^L} - V^A_{r^H}) + (r^H - r^L)\,V^W_{r^H} > 0.$$

Both value functions are bounded, so their difference is bounded, and therefore

$$\widehat{V}_L(t) - V^W_{r^H}(t) = \int_t^\infty e^{-\int_t^{t'}(r^L+\theta(u)+\lambda)\,du}\Big[\theta(t')\big(V^A_{r^L}(t') - V^A_{r^H}(t')\big) + (r^H - r^L)V^W_{r^H}(t')\Big]dt' > 0.$$

By Lemma 3, $R - x_\iota(\rho(t)) \leqslant V^W_{r^H}(t) < \widehat{V}_L(t) \leqslant V^W_{r^L}(t)$. So fabricating is strictly worse than waiting for a patient respondent, and $f_{r^L}(t) = 0$. □

Proof of Lemma 5. By (9), $\Psi[\rho] = x_\iota^{-1}\big(R - V^W_{r^H}(\cdot;\rho)\big)$. Let $Q(t) := e^{\Delta(t)}\int_t^\infty e^{-\Delta(t')}\theta(t')\,dt'$. Since

$$\int_0^\infty \exp\!\left(-r^H s - \int_0^s \theta(u+t)\,du\right)(\theta(s+t)+r^H)\,ds = 1,$$

we have

$$Q(t) = 1 - r^H\int_0^\infty \exp\!\left(-r^H s - \int_0^s \theta(u+t)\,du\right)ds,$$

and bounding the last integral between $1/(r^H+\bar\theta)$ and $1/(r^H+\underline{\theta})$ gives

$$\frac{\underline{\theta}}{r^H+\underline{\theta}} \leqslant Q(t) \leqslant \frac{\bar\theta}{r^H+\bar\theta}. \tag{24}$$

Delay bound. For $r = r^H$, the optimal delay $\sigma(t)-t$ in (5) lies in $[0,\bar\tau]$ for some $\bar\tau$ independent of t and ρ. Indeed, fix any $\tau_0 > 0$. Since k is strictly increasing and $k(0) \geqslant 0$, $k(u) \geqslant k(\tau_0) > 0$ for $u \geqslant \tau_0$, so the objective at delay $\tau = \sigma(t) - t \geqslant \tau_0$ is at most

$$e^{-r^H\tau}R - k(\tau_0)\int_{\tau_0}^{\tau} e^{-r^H u}\,du,$$

which falls below $R - x_c(\widehat{\rho}) \leqslant V^A_{r^H}$ for τ large.

Range. Since $R - x_c(\widehat{\rho}) \leqslant V^A_{r^H} \leqslant R$, (9) gives $(R - x_c(\widehat{\rho}))\,Q \leqslant V^W_{r^H} \leqslant R\,Q$. By (24),

$$\frac{r^H}{r^H+\bar\theta}R \;\leqslant\; R - V^W_{r^H}(t;\rho) \;\leqslant\; R - \frac{\underline{\theta}}{r^H+\underline{\theta}}(R - x_c(\widehat{\rho})),$$

so $\Psi[\rho](t) \in [\underline{\rho}, \bar{\rho}]$. The lower end exceeds $x_\iota(0)$ because $r^H > \bar{\theta} x_\iota(0)/(R - x_\iota(0))$, so $\underline{\rho} > 0$. The upper end is below R, so $\bar{\rho} < x_\iota^{-1}(R) = \widehat{\rho}$.

Lipschitz bound. By the maximum theorem, $V^A_{r^H}(\cdot;\rho)$ is continuous whenever ρ is. Differentiating (9), $V^W_{r^H}(\cdot;\rho)$ is then continuously differentiable, with

$$\dot{V}^W_{r^H}(t;\rho) = (r^H + \theta(t))\, V^W_{r^H}(t;\rho) - \theta(t)\, V^A_{r^H}(t;\rho),$$

as in (8). Both terms on the right lie in $[0, \bar{\theta}R]$, because $(r^H+\theta(t))V^W_{r^H}(t;\rho) \leqslant (r^H+\bar{\theta})\, R\, Q(t) \leqslant \bar{\theta}R$ by (24). Thus $|\dot{V}^W_{r^H}| \leqslant \bar{\theta}R$, and

$$|\dot{\Psi}[\rho](t)| = \frac{|\dot{V}^W_{r^H}(t;\rho)|}{x_\iota'(\Psi[\rho](t))} \leqslant \beta_1.$$

Therefore $\Psi[\rho] \in \mathcal{K}$.

Compactness. Equip $\mathbb{R}^{\mathbb{R}_+}$ with the product topology, under which it is a locally convex space. The set $\mathcal{K}$ is convex. It is closed, because pointwise limits of β_1-Lipschitz functions with values in $[\underline{\rho}, \bar{\rho}]$ are again such functions. It is contained in $[\underline{\rho}, \bar{\rho}]^{\mathbb{R}_+}$, which is compact by Tychonoff's theorem. Hence $\mathcal{K}$ is compact.

Continuity. Let $\rho^\nu \to \rho$ in $\mathcal{K}$. Because $\mathcal{K}$ is uniformly bounded and equicontinuous, the product topology on $\mathcal{K}$ coincides with the topology of uniform convergence on compact sets, which is metrizable; hence sequential continuity suffices, and the convergence is uniform on compact sets. By the maximum theorem, $V^A_{r^H}(t';\rho^\nu) \to V^A_{r^H}(t';\rho)$ for every t'. Since these functions are bounded by R, dominated convergence gives $\Psi[\rho^\nu](t) \to \Psi[\rho](t)$ for every t, so Ψ is continuous on $\mathcal{K}$.

By the Schauder–Tychonoff fixed point theorem, Ψ has a fixed point $\rho^* \in \mathcal{K}$. It is continuously differentiable because $\rho^* = \Psi[\rho^*]$. □

Proof of Lemma 6. The derivative of the objective in (5) with respect to the delay $\tau = \sigma(t) - t \geqslant 0$ is $e^{-r^H\tau}\big[\Phi_{r^H}(t+\tau) - k(\tau)\big]$, with Φ_r as in (19). Since $|\dot{\rho}| \leqslant \beta_1$ and (12) holds, $\Phi_{r^H} \leqslant -r^H(R - x_c(\widehat{\rho})) + \bar{x}_c' L < 0 \leqslant k$. Hence the objective is strictly decreasing in τ, and $\tau = 0$ is optimal. □

Proof of Lemma 7. We first define the constant in Assumption 2:

$$M := r^L \bar{x}_c' \beta_1 + \bar{x}_c'' \beta_1^2 + \bar{x}_c' \beta_2, \qquad \beta_2 := \frac{1}{\underline{x}_\iota'}\Big(\sup_t |\dot{\theta}(t)|\, R + (r^H + \bar{\theta})\, \bar{\theta} R + \bar{\theta}\, \bar{x}_c' \beta_1 + \bar{x}_\iota'' \beta_1^2\Big), \tag{25}$$

where $\beta_1 = \bar{\theta} R / \underline{x}_\iota'$, $\bar{x}_c'' := \max_{[0,\widehat{\rho}]} |x_c''|$, and $\bar{x}_\iota'' := \max_{[0,\widehat{\rho}]} |x_\iota''|$. We show below that $|\ddot{\rho}^*| \leqslant \beta_2$ and that M bounds $|\dot{\Phi}_{r^L}|$.

Bounds on ρ^*. By Lemma 5, $|\dot{\rho}^*| \leqslant \beta_1$. Under (12), $V^A_{r^H}(t; \rho^*) = R - x_c(\rho^*(t))$, which is C^1 with derivative bounded by $\bar{x}_c' \beta_1$. Write V^W and V^A for $V^W_{r^H}(\cdot; \rho^*)$ and $V^A_{r^H}(\cdot; \rho^*)$. Recall that $x_\iota(\rho^*) = R - V^W$ and $\dot{V}^W = (r^H + \theta) V^W - \theta V^A$. Hence $\dot{V}^W$ is C^1, with

$$\ddot{V}^W = \dot{\theta}\,(V^W - V^A) + (r^H + \theta)\,\dot{V}^W - \theta\,\dot{V}^A.$$

Using $|V^W - V^A| \leqslant R$ (both lie in $[0, R]$), $|\dot{V}^W| \leqslant \bar{\theta} R$, and $|\dot{V}^A| \leqslant \bar{x}_c' L$, we obtain

$$|\ddot{V}^W| \leqslant \sup_t |\dot{\theta}(t)|\, R + (r^H + \bar{\theta})\, \bar{\theta} R + \bar{\theta}\, \bar{x}_c' L.$$

Differentiating $x_\iota'(\rho^*)\, \dot{\rho}^* = -\dot{V}^W$ gives

$$\ddot{\rho}^* = -\frac{\ddot{V}^W + x_\iota''(\rho^*)\, (\dot{\rho}^*)^2}{x_\iota'(\rho^*)},$$

so ρ^* is C^2 and $|\ddot{\rho}^*| \leqslant \beta_2$.

Uniqueness of the patient type's solution. Let Φ_{r^L} be as in (19) with $\rho = \rho^*$. Then Φ_{r^L} is C^1, with

$$\dot{\Phi}_{r^L} = r^L x_c' \dot{\rho}^* - x_c''(\dot{\rho}^*)^2 - x_c' \ddot{\rho}^*,$$

so $|\dot{\Phi}_{r^L}| \leqslant M < \inf k'$. The derivative of the objective in (5) with respect to the delay τ is $e^{-r^L \tau}\big[\Phi_{r^L}(t + \tau) - k(\tau)\big]$, and the bracket is strictly decreasing in τ because its derivative is $\dot{\Phi}_{r^L} - k' < 0$. Hence the objective is single-peaked in τ. As in the proof of Lemma 5, the maximizer lies in a compact interval of delays, so a solution exists. It is unique: $\tau = 0$ if $\Phi_{r^L}(t) \leqslant k(0)$, and otherwise the unique τ at which the bracket vanishes.

Properties of $\sigma^*_{r^L}$. Continuity follows from the maximum theorem, since the solution is unique. Where $\sigma^*_{r^L}(t) > t$, the first-order condition $\Phi_{r^L}(\sigma^*_{r^L}(t)) = k(\sigma^*_{r^L}(t) - t)$ holds. Its derivative in $\sigma^*_{r^L}(t)$ is $\dot{\Phi}_{r^L} - k' < 0$, so the implicit function theorem gives

$$\dot{\sigma}^*_{r^L} = \frac{k'}{k' - \dot{\Phi}_{r^L}}.$$

Because $|\dot{\Phi}_{r^L}| < k'$, the denominator lies in $(0, 2k')$, so $\dot{\sigma}^*_{r^L} > 1/2$. Strict monotonicity follows from Lemma 2. □

Proof of Theorem 1. Recall that $N_H(t)$ and $N_L(t)$ are the probabilities that at time t the respondent is silent, has not yet thought of an answer, and is impatient or patient, respectively; and that G^*_L in (13) is the density of the patient type's well-considered replies under σ^*.

By Lemma 5, $\rho^* \in \mathcal{K}$ is continuously differentiable. Since (12) holds, Lemma 6 gives $\sigma^*_{r^H}(t) = t$. Given ρ^*, the reply times σ^* are optimal by construction, and $\sigma^*_{r^L}$ is strictly increasing by Lemma 2. Because $\rho^* = \Psi[\rho^*]$, the impatient type is indifferent between fabricating and waiting at every t by (10), so any intensity $f_{r^H}(t)$ is optimal for her. By Lemma 4, $f_{r^L} \equiv 0$ is optimal for the patient type.

Since $N^*_H(t) > 0$, the intensity (15) is finite. Substituting it into (3) gives $\dot{N}_H = \lambda N_L - \theta N_H/(1-\rho^*) - \rho^* G^*_L/(1-\rho^*)$, whose solution with $N_H(0) = q$ is (14). Hence the induced probabilities are $N_H = N^*_H$ and $N_L(t) = (1-q)e^{-\Theta(t)-\lambda t}$, and the density of well-considered replies is $G = \theta N^*_H + G^*_L > 0$. By construction $F = f_{r^H} N^*_H = \rho^* G/(1-\rho^*)$, so the belief path that (4) assigns to (σ^*, f) is $F/(F+G) = \rho^*$. Thus (σ^*, f, ρ^*) is a PBE.

Independence of q, λ, and r^L follows because none of them enters (5) for $r = r^H$ or (11). □

Proof of Corollary 1. Delay bound. By Lemma 7, a patient respondent whose answer arrives at s either replies at once or delays by the unique $\tau > 0$ with $\Phi_{r^L}(s+\tau) = k(\tau)$, where Φ_{r^L} is as in (19) with $\rho = \rho^*$. Since $|\Phi_{r^L}| \leqslant r^L R + \bar{x}'_c L$ and $k(\tau) \geqslant k(0) + \tau \inf k' \geqslant \tau \inf k'$, we get $\tau \leqslant \bar{d}$. Hence $s \leqslant \sigma^*_{r^L}(s) \leqslant s + \bar{d}$.

Abundance. By Lemma 5, $\rho^* \leqslant \bar{\rho}$, so $\rho^*/(1-\rho^*) \leqslant \bar{\rho}/(1-\bar{\rho})$ and $1/(1-\rho^*) \leqslant 1/(1-\bar{\rho})$. In (14), change variables $u = \sigma^*_{r^L}(s)$ in the integral of $\rho^* G^*_L/(1-\rho^*)$; the Jacobian $1/\dot{\sigma}^*_{r^L}$ in

G_L^* cancels. Using $C(\sigma_{rL}^*(s)) - C(s) \leqslant \bar{\theta}\bar{d}/(1-\bar{\rho})$ and $(\sigma_{rL}^*)^{-1}(t) \leqslant t$,

$$\int_0^t e^{C(u)} \frac{\rho^*(u)}{1-\rho^*(u)} G_L^*(u)\, du = \int_0^{(\sigma_{rL}^*)^{-1}(t)} e^{C(\sigma_{rL}^*(s))} \frac{\rho^*(\sigma_{rL}^*(s))}{1-\rho^*(\sigma_{rL}^*(s))} \theta(s) N_L(s)\, ds$$
$$\leqslant \frac{\bar{\rho}}{1-\bar{\rho}} \exp\Big(\frac{\bar{\theta}\,\bar{d}}{1-\bar{\rho}}\Big) \int_0^t e^{C(s)} \theta(s) N_L(s)\, ds.$$

Hence

$$e^{C(t)} N_H^*(t) \geqslant q + \int_0^t e^{C(s)} N_L(s) \Big(\lambda - \frac{\bar{\rho}}{1-\bar{\rho}} \exp\Big(\frac{\bar{\theta}\,\bar{d}}{1-\bar{\rho}}\Big)\theta(s)\Big) ds \geqslant q > 0$$

under (16).

□

Proof of Proposition 1. By Lemma 5, Ψ maps $\mathcal{K}$ into itself, where

$$\Psi[\rho](t) = x_\iota^{-1}\left(R - e^{\Delta(t)} \int_t^\infty e^{-\Delta(t')}\, \theta(t')\, V_{rH}^A(t';\rho)\, dt'\right).$$

Case 1: $\dot{\theta}(t) > 0$ for all t. We show that if ρ is nonincreasing then $\Psi[\rho]$ is strictly decreasing.

Claim 2. *If $\dot{\theta}(t) > 0$, then for each nonincreasing $\rho \in \mathcal{K}$, $\Psi[\rho]$ is strictly decreasing in t.*

Proof. Since

$$V_{rH}^A(t;\rho) = \max_{\tau \geqslant 0} \Big\{ e^{-r^H \tau} (R - x_c(\rho(\tau + t))) - \int_0^\tau e^{-r^H s} k(s)\, ds \Big\},$$

nonincreasingness of ρ implies that V_{rH}^A is nondecreasing in t, with values in $(R - x_c(\widehat{\rho}), R)$. Then

$$\dot{\Psi}[\rho](t) = -\frac{1}{x_\iota'(\Psi[\rho](t))} \Bigg(e^{\Delta(t)} \left[\int_t^\infty e^{-\Delta(t')}\, \theta(t')\, V_{rH}^A(t';\rho)\, dt' \right] (r^H + \theta(t))$$
$$- \theta(t)\, V_{rH}^A(t;\rho) \Bigg). \tag{26}$$

Using that $V^A_{r^H}(t';\rho)$ is nondecreasing in t',

$$\int_t^\infty e^{-\Delta(t')}\,\theta(t')\,V^A_{r^H}(t';\rho)\,dt' \geqslant V^A_{r^H}(t;\rho)\int_t^\infty e^{-\Delta(t')}\,\theta(t')\,dt'$$
$$= V^A_{r^H}(t;\rho)\left[e^{-\Delta(t)} - r^H\int_t^\infty e^{-\Delta(t')}\,dt'\right].$$

Substituting into (26),

$$\dot\Psi[\rho](t) \leqslant -\frac{V^A_{r^H}(t;\rho)\,r^H}{x'_\iota(\Psi[\rho](t))}\,e^{\Delta(t)}\left[e^{-\Delta(t)} - (r^H+\theta(t))\int_t^\infty e^{-\Delta(t')}\,dt'\right]$$
$$< -\frac{V^A_{r^H}(t;\rho)\,r^H}{x'_\iota(\Psi[\rho](t))}\,e^{\Delta(t)}\left[e^{-\Delta(t)} - (r^H+\theta(t))\int_t^\infty e^{-\Delta(t)-(r^H+\theta(t))(t'-t)}\,dt'\right] = 0,$$

where the strict inequality uses $\Delta(t') - \Delta(t) > (r^H+\theta(t))(t'-t)$ for $t' > t$, which holds because θ is strictly increasing. □

Let $\mathcal{K}^-$ be the set of nonincreasing functions in $\mathcal{K}$. It is convex and closed in the product topology, hence compact. By Claim 2 and Lemma 5, $\Psi(\mathcal{K}^-) \subset \mathcal{K}^-$, and the Schauder–Tychonoff theorem yields a fixed point $\rho^* \in \mathcal{K}^-$. By Claim 2 applied to ρ^*, $\rho^* = \Psi[\rho^*]$ is strictly decreasing.

Case 2: $\dot\theta(t) < 0$. Parallel to Case 1, with monotonicity reversed.

Case 3: θ *constant.* Setting $\rho(t) = \rho$, we have $V^A_{r^H}(t;\rho) = R - x_c(\rho)$, and

$$\Psi[\rho] = x_\iota^{-1}\left(R - \frac{\theta}{\theta + r^H}(R - x_c(\rho))\right).$$

Brouwer's theorem yields a constant fixed point. □

Proof of Lemma 8. Since ρ is monotone and bounded, it converges to some $\rho(\infty)$. Since θ is monotone and bounded, $\theta(\infty)$ exists. Because ρ is monotone, $V^A_{r^H}(t;\rho)$ is monotone in t and bounded, so it converges to some V_∞. By dominated convergence in (9), $V^W_{r^H}(t;\rho) \to \theta(\infty)V_\infty/(r^H+\theta(\infty))$. Hence

$$\dot V^W_{r^H}(t;\rho) = (r^H+\theta(t))V^W_{r^H}(t;\rho) - \theta(t)V^A_{r^H}(t;\rho) \to 0,$$

and so $\dot{\rho}(t) = -\dot{V}^W_{r^H}(t;\rho)/x'_\iota(\rho(t)) \to 0$.

It follows that, for t large, $|x'_c(\rho(s))\,\dot{\rho}(s)| < r^H(R - x_c(\widehat{\rho}))$ for all $s \geqslant t$. The derivative of the objective in (5) for $r = r^H$ is then strictly negative at every delay, because $k \geqslant 0$. Hence $\sigma_{r^H}(t) = t$ and $V^A_{r^H}(t;\rho) = R - x_c(\rho(t))$ for large t, so $V_\infty = R - x_c(\rho(\infty))$. Taking limits in $\rho(t) = x_\iota^{-1}(R - V^W_{r^H}(t;\rho))$ yields (17). □

Proof of Lemma 9. Let $\|\cdot\|$ denote the supremum norm. For every t', since $|\max_\tau a(\tau) - \max_\tau b(\tau)| \leqslant \sup_\tau |a(\tau) - b(\tau)|$ and $e^{-r^H\tau} \leqslant 1$,

$$\left|V^A_{r^H}(t';\rho) - V^A_{r^H}(t';\omega)\right| \leqslant \sup_{\tau\geqslant 0}\left|x_c(\rho(t'+\tau)) - x_c(\omega(t'+\tau))\right| \leqslant \bar{x}'_c\,\|\rho - \omega\|.$$

Hence, by (9) and with Q as in the proof of Lemma 5,

$$|V^W_{r^H}(t;\rho) - V^W_{r^H}(t;\omega)| \leqslant Q(t)\,\bar{x}'_c\,\|\rho - \omega\| \leqslant \frac{\bar{\theta}}{r^H + \bar{\theta}}\,\bar{x}'_c\,\|\rho - \omega\|$$

by (24), and $|\Psi[\rho](t) - \Psi[\omega](t)| \leqslant |V^W_{r^H}(t;\rho) - V^W_{r^H}(t;\omega)|/\underline{x}'_\iota \leqslant \kappa\|\rho - \omega\|$. The continuous belief paths with values in $[0,\widehat{\rho}]$ form a complete metric space under $\|\cdot\|$, and Ψ maps this space into itself by Lemma 5. If $\kappa < 1$, the contraction mapping theorem gives a unique fixed point. □

Proof of Lemma 10. Recall that the derivative of the objective in (5) with respect to the delay $\tau = \sigma(t) - t$ is $e^{-r\tau}\big[\Phi_r(t+\tau) - k(\tau)\big]$. If $\sigma_r(t) = t$ is optimal for every t, this derivative is nonpositive at $\tau = 0$, that is, $\Phi_r(t) \leqslant k(0)$ for every t. Conversely, if $\Phi_r(s) \leqslant k(0)$ for every s, then for every $\tau > 0$ the derivative is at most $e^{-r\tau}[k(0) - k(\tau)] < 0$, because k is strictly increasing; so the objective is strictly decreasing in τ and $\sigma_r(t) = t$. □

Proof of Theorem 2. By Claim 2 applied to $\rho^* = \Psi[\rho^*]$, $\dot{\rho}^*(t) < 0$ for every t. Fix any $\widehat{t}$. By Theorem 1, ρ^* does not depend on r^L. Hence, for

$$r^L < \frac{x'_c(\rho^*(\widehat{t}))\,|\dot{\rho}^*(\widehat{t})|}{R - x_c(\rho^*(\widehat{t}))},$$

we have $\Phi_{r^L}(\widehat{t}) > 0 = k(0)$, where Φ_r is defined in (19) with $\rho = \rho^*$. The derivative of the

objective in (5) at $\tau = 0$ for a patient respondent whose answer arrives at $\widehat{t}$ is therefore strictly positive, so $\sigma_{rL}(\widehat{t}) > \widehat{t}$. Since Φ_{rL} is continuous, the same holds on a neighborhood of $\widehat{t}$. Patient respondents with an answer at such times exist on the equilibrium path, because $N_L(t) > 0$ for every t. □

Proof of Proposition 2. Suppose such a PBE exists. By (10), its suspicion path satisfies $\rho = \Psi[\rho]$, so $\rho \in \mathcal{K}$ by Lemma 5. Hence $\rho \geqslant \underline{\rho} > 0$, and $\sigma_{rH}(t) = t$ by Lemma 6.

Bayes' rule gives $f_{rH} N_H = \rho\,(\theta N_H + G_L)/(1-\rho)$. With $\lambda = 0$, (3) becomes $\dot{N}_H = -\theta N_H/(1-\rho) - \rho\, G_L/(1-\rho)$. Solving,

$$N_H(t) = e^{-C(t)}\Big[q - \int_0^t e^{C(s)} \frac{\rho(s)}{1-\rho(s)}\, G_L(s)\, ds\Big], \qquad C(t) := \int_0^t \frac{\theta(s)}{1-\rho(s)}\, ds.$$

We show that the integral diverges. By Lemma 7, applied to the fixed point ρ, σ_{rL} is differentiable wherever it delays, so we can change variables $s = \sigma_{rL}(u)$; the Jacobian in G_L cancels. We also use $\sigma_{rL}(u) \geqslant u$, the monotonicity of C, $C(u) \geqslant \Theta(u)/(1-\underline{\rho})$, and $N_L(u) = (1-q)e^{-\Theta(u)}$. This gives

$$\begin{aligned}\int_0^{\sigma_{rL}(T)} e^{C(s)} \frac{\rho(s)}{1-\rho(s)}\, G_L(s)\, ds \;&\geqslant\; (1-q)\,\frac{\underline{\rho}}{1-\underline{\rho}} \int_0^T \exp\Big(\frac{\underline{\rho}\,\Theta(u)}{1-\underline{\rho}}\Big)\theta(u)\, du \\ &= (1-q)\Big(\exp\Big(\frac{\underline{\rho}\,\Theta(T)}{1-\underline{\rho}}\Big) - 1\Big),\end{aligned}$$

which tends to infinity as $T \to \infty$, since $\Theta(T) \geqslant \underline{\theta} T$. Hence $N_H(t) < 0$ for some finite t, a contradiction. □

Proof of Lemma 11. By the derivatives computed in the proof of Lemma 1, $x'_G/x'_F = (1-\bar{w})/\bar{w} = \ell$, where $\ell = \frac{\rho}{1-\rho} \cdot \frac{1-w}{w}$. □

Proof of Proposition 3. By (1) and (2), the functions x_F, x_G, x_c, and x_ι depend on ρ and w only through $\bar{w}(\rho; w)$. Since $\bar{w}(\rho'(t); w') = \bar{w}^*(t) = \bar{w}(\rho^*(t); w)$, we have

$$x_i(\rho'(t); w') = x_i(\rho^*(t); w) \qquad (i = F, G, c, \iota)$$

for every t. In particular, every reply faces the same investigation probabilities under w' as under w. Because $\bar{w}(\cdot; w')$ is smooth with a smooth inverse, ρ' is continuously differentiable, and $x_\iota(\rho'(t); w') = x_\iota(\rho^*(t); w) \leqslant R$ gives $\rho'(t) \in [0, \widehat{\rho}_{w'}]$.

Problem (5) depends on the belief path only through $t \mapsto x_c(\rho(t))$, which is the same under (ρ', w') as under (ρ^*, w). Hence the two problems have the same solutions σ^* and the same values V_r^A, and by (9) the same V_{rH}^W. The impatient type is therefore indifferent at every t under w', since $R - x_\iota(\rho'(t); w') = R - x_\iota(\rho^*(t); w) = V_{rH}^W(t)$, so ρ' is a fixed point of Ψ under w'. The patient type does not fabricate, by Lemma 4.

The probabilities N_L and the density G_L^* depend only on σ^* and are unchanged. By the abundance assumption, (14) with ρ^* replaced by ρ' stays positive. The intensity in (15) with that replacement is therefore finite and generates ρ' through (4), exactly as in the proof of Theorem 1. The last claim follows from uniqueness. □